Topic 3 – Cell Structure, Reproduction and Development

Candidates will be assessed on their ability to:

3.1	know that all living organisms are made of cells, sharing some common features
3.2	understand how the cells of multicellular organisms are organised into tissues, tissues into organs, and organs into organ systems
3.3	(i) know the ultrastructure of eukaryotic cells, including nucleus, nucleolus, ribosomes, rough and smooth endoplasmic reticulum, mitochondria, centrioles, lysosomes and Golgi apparatus (ii) understand the function of the organelles listed in (i)
3.4	understand the role of the rough endoplasmic reticulum (rER) and the Golgi apparatus in protein transport within cells, including their role in the formation of extracellular enzymes
3.5	(i) know the ultrastructure of prokaryotic cells, including cell wall, capsule, plasmid, flagellum, pili, ribosomes and circular DNA (ii) knderstand the function of the structures listed in (i)
3.6	be able to recognise the organelles in **3.3** from electron microscope (EM) images
3.7	(i) know how magnification and resolution can be achieved using light and electron microscopy (ii) understand the importance of staining specimens in microscopy
3.8	**CORE PRACTICAL 5** **(i) use a light microscope to make observations and labelled drawings of suitable animal cells** **(ii) use a graticule with a microscope to make measurements and understand the concept of scale**
3.9	(i) know that a locus is the location of genes on a chromosome (ii) understand the linkage of genes on a chromosome
3.10	understand the role of meiosis in ensuring genetic variation through the production of non-identical gametes as a consequence of independent assortment of chromosomes in metaphase I and crossing over of alleles between chromatids in prophase I *Names of the stages of prophase are not required.*
3.11	understand how mammalian gametes are specialised for their functions (including the acrosome in sperm and the zona pellucida in the egg cell)
3.12	know the process of fertilisation in mammals, including the acrosome reaction, the cortical reaction and the fusion of nuclei
3.13	know the process of fertilisation in flowering plants, starting with the growth of a pollen tube and ending with the fusion of nuclei
RECOMMENDED ADDITIONAL PRACTICAL **Investigate factors affecting the growth of pollen tubes.**	

3.15	**CORE PRACTICAL 6** **Prepare and stain a root tip squash to observe the stages of mitosis.**
3.16	be able to calculate mitotic indices
3.17	(i) understand what is meant by the terms *stem cell*, *pluripotent* and *totipotent*, *morula* and *blastocyst* (ii) be able to discuss the ways in which society uses scientific knowledge to make decisions about the use of stem cells in medical therapies
3.18	understand how cells become specialised through differential gene expression, producing active mRNA, leading to the synthesis of proteins which, in turn, control cell processes or determine cell structure in animals and plants
3.19	understand how one gene can give rise to more than one protein through post-transcriptional changes to messenger RNA (mRNA)
3.20	(i) understand how phenotype is the result of an interaction between genotype and the environment (ii) know how epigenetic modification, including DNA methylation and histone modification, can alter the activation of certain genes (iii) understand how epigenetic modifications can be passed on following cell division
3.21	understand how some phenotypes are affected by multiple alleles for the same gene, or by polygenic inheritance, as well as the environment, and how polygenic inheritance can give rise to phenotypes that show continuous variation